

Experiment 2: Exploratory Data Analysis (EDA) on a Real-World Business Dataset

Questions & Answers

Q1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?

- **Definition:** EDA is the process of analyzing datasets through summary statistics and visual representations to understand their structure, distributions, and underlying relationships.
- **Why before Machine Learning:** It helps detect anomalies, handle missing values, uncover feature patterns, and validate assumptions required for reliable predictive modeling.

Q2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.

- **Univariate Analysis:** Analyzes a single variable at a time (e.g., studying the histogram of employee ages).
- **Bivariate Analysis:** Examines the relationship between two variables (e.g., scatter plot of house area vs. price).
- **Multivariate Analysis:** Explores interactions among three or more variables simultaneously (e.g., analyzing sales across regions colored by product categories).

Q3. What insights can be obtained from a correlation heatmap?

- **Feature Relationships:** Identifies strong linear relationships (positive or negative) between continuous numerical variables.
- **Multicollinearity Detection:** Highlights redundant or highly collinear independent variables to avoid overfitting in regression models.

Q4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

- **Histogram:** Visualizes the frequency distribution, central tendency, and skewness of a single continuous variable.
- **Box Plot:** Summarizes data dispersion using quantiles and identifies statistical outliers.
- **Scatter Plot:** Displays relationship trends and cluster groupings between two continuous variables.

Q5. How can EDA help identify data quality issues before analysis?

- Flags missing or null values in dataset attributes.
- Spotlights extreme values or data entry errors via box plots and scatter plots.
- Uncovers inconsistent categorical entries (e.g., duplicate names with varied casing) and severe class imbalances.

Q6. Why is correlation important in predictive analytics? Can correlation imply causation?

- **Importance:** Helps select key features that strongly influence the target outcome, improving prediction quality.

- **Causation: No, correlation does not imply causation.** Two variables may move together due to a third confounding factor or purely by coincidence, not necessarily because one directly causes the other.

Q7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

- **Chosen Visualization: Box Plot or Grouped Bar Plot (Count/Mean).**
- **Justification:** Box plots clearly show how a numerical metric's median, spread, and outliers vary across distinct categories.

Q8. What business insights can be derived from the Netflix (or Superstore/HR Analytics) dataset through EDA?

- **Netflix Example:** Reveals shift trends from movies to multi-season TV shows, dominant content ratings (e.g., TV-MA), and top content-producing regions.
- **Superstore Example:** Identifies non-profitable product sub-categories, high-performing regions, and seasonality in annual sales.

Q9. How does EDA contribute to feature selection and model building?

- Helps drop non-informative or redundant features with low variance or extreme correlation.
- Guides feature transformation decisions (e.g., log transforms for skewed features).
- Informs categorical encoding strategies based on category frequency distributions.

Q10. What challenges might arise while performing EDA on large-scale real-world datasets?

- High memory usage and computational slowness when loading large datasets into RAM.
- Overplotting in scatter plots when rendering millions of data points.
- Managing noisy, incomplete, or high-cardinality categorical data.